

Phase 4: Statistics & Business Applications (Month 3)

Business Scenario–Based Exercise Questions

Module Goal

Apply statistical concepts to real-world business decisions using practical scenarios from retail, HR, marketing, and finance.

Section 1: Descriptive vs. Inferential Statistics

Exercise 1: Retail Store Performance

A retail chain collected monthly sales data from 12 stores across different cities.

Store Sales (\$000): A-120, B-135, C-128, D-142, E-150, F-138, G-145, H-132, I-140, J-155, K-148, L-130

Questions:

Calculate mean, median, mode, and standard deviation

Ans : mean = $120+135+128+142+150+138+145+132+140+155+148+130$
 $=1663/12=138.58$

Mean is 138.58

Median = 12 is even number

$= n+1/2$

$=12+1/2$

$=13/2$

$=6.5$

$=138+140=278/2=139$

Median is 139

Median is between 6th and 7th term that is between 138 and 145

3. mode : mode is 0 no value is there that is repeated

4. sd : 138.58

X

x	(x-m)	(x-m) ²
120	-18.58	345.22
135	-3.58	12.82
128	-10.58	111.94
142	3.42	11.70
150	11.42	130.42
138	-0.58	0.34
145	6.42	41.22
132	-6.58	43.30
140	1.42	2.02
155	16.42	269.62
148	9.42	88.74
130	-8.58	73.62

$$(x-m)^2 = 1130.96$$

$$Sd = \sqrt{(1130.96/12)}$$

$$sd = \sqrt{94.25}$$

$$sd=9.71$$

standard deviation = 9.71

2. Interpret the standard deviation.

The standard is 9.71 and the mean is 138.58 the spreadness between the data points is high but is it not extreme so we can the distribution of data is within the range
Sd is 9.71

3. Identify descriptive and inferential statistics.

Ans : mean , median , mode , sd descriptive statistics

Interpret the sd based on population or mean for the population of 12 cities is the inferential statistics

4. Explain whether estimating sales for all stores is inferential analysis.

Ans : yes we will called it is an inferential statistics because estimating for all stores is estimation of population but in this case no specific info is given

Section 2

Section 2: Mean, Median, Mode & Standard Deviation

Exercise 2: Employee Salary Analysis

Salaries (\$): 40,000; 42,000; 43,000; 44,000; 45,000; 46,000; 47,000; 120,000

Questions:

Calculate mean, median, range, and standard deviation

$$\text{Mean} = 40000 + 42000 + 43000 + 44000 + 45000 + 46000 + 47000 + 120000$$

$$= \text{total} / 8$$

$$= 427000 / 8$$

$$= 53375$$

$$\text{Median} = 8 + 1/2$$

$$= 9/2$$

$$= 4.5$$

4th and 5th term

$$= 44000 + 45000$$

$$= 44500$$

Median is 44500

Range = highest – lowest value

$$= 120000 - 40000 = 80000$$

Sd : find mean

$$m = 40000 + 42000 + 43000 + 45000 + 44000 + 46000 + 47000 + 120000 = 427000$$

$$n = 8$$

$$m = 427000 / 8 = 53375$$

x	(x-m)	(x-m) ²
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40000	-13375	178,890,625
42000	-11375	129,390,625
43000	-10375	107,640,625
44000	-9375	87,890,625
45000	-8375	70,140,625
46000	-7375	54,390,625
47000	-6375	40,640,625
120000	66625	4,438,890,625

$$(x-m)^2 = 5,107,875,000$$

$$Sd = \sqrt{5107875000/8}$$

$$Sd = \sqrt{638484375}$$

$$Sd = 25268$$

2. Explain why mean is higher than median.

Ans : The mean is higher than the median because the CEO salary (\$120,000) is an outlier that pulls the average upward mean is always lower the median

3. Which measure better represents salaries?

Ans : The median better represents salaries because it is less affected by extreme values and better reflects the typical employee salary. Median is never ever impacted by outlier it is also the best for central tendency

4. Analyze effect of removing CEO salary.

Ans : since the ceo salary is outlier After removing the CEO salary, the mean becomes \$43,857.14 and the median becomes \$44,000.00. The salaries appear more balanced and representative of regular employees.

5. Explain impact of outliers.

Ans : Outliers increase the mean and standard deviation, making the data appear more spread out and potentially misleading

Section 3: Probability & Business Risk

Exercise 3: Loan Approval Risk Analysis

Low Risk: 2%, Medium Risk: 8%, High Risk: 20%
Applications: 500 Low, 300 Medium, 200 High

Questions:

1. Calculate expected defaults.
. Expected defaults: Low Risk = 10, Medium Risk = 24, High Risk = 40.
Low risk 2 percent of 500 =10
Medium risk 8 percent of 300=24
High risk 20 percent of 200=40

2. Calculate total defaults.

Ans : total defaulters = 10+24+40 = 74 people from 1000 applications

3. Identify highest-risk segment.

Ans : The High Risk segment has the highest probability of default (20%).

- 4 Recommend focus area for reducing defaults.

Ans : The company should focus on reducing defaults in the High Risk segment through stricter approval criteria and monitoring.

- 5.Explain business importance of probability.

Ans : . Probability helps businesses estimate risks, plan strategies, reduce losses, and make informed decisions.

Section 4: Hypothesis Testing

Exercise 4: Manufacturing Quality Control

A packaging machine should fill bottles with 500 ml. Sample mean = 492 ml; SD = 12 ml.

Questions:

1. Define H_0 and H_1

Ans : Null Hypothesis (H_0): The machine fills bottles with 500 ml on average. Alternative Hypothesis (H_1): The machine does not fill bottles with 500 ml on average.

2. Decide whether investigation is required.

Ans : Yes. Since the sample mean is 492 ml, which is noticeably below the target of 500 ml, an investigation is required.

3. Explain Type I and Type II errors.

Ans : . Type I Error: Concluding the machine is faulty when it is actually working correctly. Type II Error: Concluding the machine is working correctly when it is actually faulty.

Type 1 error : false positive : machine is working but we assume it is faulty

Type 2 error : true negative : machine is not working and is default but we assume it is working properly

4. Discuss operational impact.

Ans : Incorrect bottle filling can lead to customer complaints, regulatory issues, financial losses, and reduced trust in product quality.

5. Explain importance of hypothesis testing.

Ans : Hypothesis testing helps organizations make objective decisions using data rather than assumptions.

Section 5: A/B Testing for Marketing Decisions

Exercise 5: Email Marketing Campaign

Version A: 5,000 emails, 350 clicks

Version B: 5,000 emails, 500 clicks

Questions:

1. Calculate CTR for both campaigns.

Ans : CTR : CTR is the click per time rate $350/5000 \times 100 = 7\%$

$500/5000 \times 100 = 10\%$

2. Identify better-performing campaign.

Ans : . Version B performed better because it achieved more clicks.

3. Calculate percentage improvement.

Ans : Percentage improvement of Version B over Version A = 42.86%.

This shows that how much better version B performed than version A

$\text{New CTR} - \text{OLD CTR} / \text{OLD CTR} \times 100$

$= \text{old ctr} = 7\% \text{ (version a)}$

New ctr = 10% version B

Version B = $10 - 7 = 3\%$

$3/7 = 0.4286$

$= 0.4286 \times 100 = 42.86\%$

4. Explain usefulness of A/B testing.

Ans : A/B testing helps businesses compare strategies and identify which version performs better based on real data.

5. Identify risks reduced through testing.

Ans : Testing reduces risks such as poor campaign performance, wasted marketing budget, and low customer engagement.

6. Suggest additional metrics.

Ans : Additional useful metrics include conversion rate, unsubscribe rate, bounce rate, revenue generated, and customer retention.

Section 6: Data-Driven Decision Making

Exercise 6: Restaurant Expansion Decision

City A: Revenue 210, Growth 2%, Competitors 15

City B: Revenue 180, Growth 6%, Competitors 7

City C: Revenue 240, Growth 1%, Competitors 20

1. Which city has highest revenue?

Ans : city C : 240

2. Which city is best for long-term growth?

Ans : City B is best for long-term growth because it has the highest growth rate (6%) and fewer competitors.

3. What additional data is required?

Ans : . Additional data required includes population size, customer demographics, rental costs, labor availability, and average customer spending.

4. Explain importance of data-driven decisions.

Ans : Data-driven decisions improve accuracy, reduce uncertainty, and help businesses allocate resources effectively.

5. Recommend a city with justification

Ans : City B is recommended because it offers strong growth potential with lower competition, increasing the chances of long-term success.